

On Stock Trading: Can a Trend Follower Expect to Win?

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Abstract

Our answer to this question is a qualified “yes.” In an idealized market with perfect liquidity, continuous trading and no transaction costs, if a trend follower is a small enough player so as to be a price-taker with adequate resources to avoid margin calls, we rigorously prove that a positive expected value of the trading gain can be guaranteed. Surprisingly, the trend-following trading strategy which we describe, called *Simultaneous Long-Short* (SLS), is model-free. That is, the trader requires neither an “internal model” for the underlying stock prices nor a behavioral model for the actions of other players in the marketplace. While there are a number of papers in the literature which demonstrate the efficacy of such model-free strategies in an empirical data context via back-testing, to our knowledge, there is a little or no literature which includes a purely theoretical rationale to explain why such methods may work well in practice. The SLS strategy, motivated by ideas in feedback control theory, has the following salient feature: Rather than use of a model of any sort to adjust the amount invested over time, profits and losses are the only factor used for adaptation. More precisely, the number of dollars invested $I(t)$ at time t is adapted as a function of the cumulative gains or losses $g(t)$ on $[0, t]$. Subsequently, the “power” of our trading scheme is demonstrated using Geometric Brownian Motion for the prices. We include a rather arbitrary time-varying drift $\mu(t)$ and a time-varying volatility $\sigma(t)$ as well. The class of admissible time variations negates the efficacy of many obvious trading strategies based on estimation of the direction in which the prices are drifting. Despite this lack predictability, we rigorously prove that the SLS trader “expects” to make money on every trade. That is, for any non-trivial price variation and any $t > 0$, the expected value of the trading gain satisfies $\mathbb{E}[g(t)] > 0$. To our knowledge, such a remarkable property, attainment of performance robustness in a framework without any model to reveal the directional movement of prices, has never been uncovered. We believe that these results may be the “tip of the iceberg” in terms of new research directions aimed at revealing a theory which explains the empirically observed success of trend-following strategies beyond the one considered here.

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I. INTRODUCTION

Up until the late eighties, the entire area of technical analysis was generally considered to be the “voodoo” of the finance world by the academic community. Early papers such as the one by Fama and Blume [1], involving back-testing using historical data and point to the statistical insignificance of profits accrued via some well-known trading rules; see also [2] similar conclusions are reached. More generally, in the book by Malkiel [3], the arguments against technical analysis, largely based on weak-form efficiency considerations, are articulated in great detail.

Suffice it to say most academics agreed that technical trend forecasts, generated by price and volume, are of questionable worth in a trading context. While some technical stock trading rule might produce impressive gains over some period of time, detractors of technical analysis typically argued that such success is an anomaly and that there is no credible theoretical foundation, based on financial theory, which makes it worthy of serious study. If a trader uses a technical trading strategy which produces superior returns over some period of time, it was argued that it would eventually become widely known and the extra “juice from the orange” will be “arbitraged away” in short order. That is, after some period of time, the strategy will cease to be efficacious.

The tide began to turn during the late eighties. For example, papers such as [4] revisited the earlier statistical analysis in [1] and arrived at a more optimistic conclusion when the data was analyzed in a different manner. On the theoretical side, papers such as [5], [6], [7] and [8] further served to increase the interest of academia in technical analysis. These papers, based on behavioral models involving considerations such as rational expectations, utility functions and equilibria, proffered possible explanations of observed marketplace phenomena and why traders might behave in a certain manner.

In the early nineties, on the heels of [4], we also see a significant renewal of interest in the use of empirical data to conduct statistical studies involving back-testing of trading algorithms. Of particular note, in [9], a rather systematic empirical study concentrated on the attainment

of superior returns via back-testing a number of rules involving entry and exit from a trade based on moving average and breakout considerations.¹ In the years to follow, a number of other statistical studies involving technical trading rules based on empirical data added fuel to the fire; e.g., see [10], [11], [12], [13] and [14]. Finally, in the nineties, further momentum for data-driven technical analysis was provided by brokerage houses that routinely published their view of markets.²

This paper represents a radical departure from both the statistical and model-based lines of research described above. In a trend-following stock-trading context, we claim that there is a rigorous theory which explains the possibility of superior returns and involves neither statistical back-testing nor behavioral modelling. In fact, surprising as this may seem, our theory uses no model at all. In an ideal market characterized by perfect liquidity, continuous trading, no transaction costs and price-taking, our claim is that sound trade management rules, based solely on adaptation of one's stock position size to gains and losses, leads to a positive expected value of the trading gain irrespective of the conditions of entry.

To provide additional perspective before our technical exposition, we begin by noting that trend following is a sub-class of technically-based strategies. It is largely centered on the notion that markets may move in one direction or another, and with proper entry and trade management rules, one may consistently profit from these movements. While trend-following strategies come in many flavors, they all tend to have some common features: First, they are direction independent. That is, the goal is to simply “catch a trend,” whether up or down. Second, they typically follow the principle of “ride the trend.” That is, a forecasted price increase is exploited by a long position and a decrease via short position. In addition, the trade is managed in such a way so

¹Note that [11] has brought the results of [9] into question. However, it nevertheless remains true that in the years following [9], this line of research gained considerable momentum.

²We do not include the multitude of “popular” technical analysis books as evidence that technical analysis is increasingly being viewed with credibility in academic circles.

that “profits run.” Finally, losses are generally cut short.³

Before getting to our specific claims, when discussing trend-following trading methods, for the purpose of this paper, it is important to make a distinction between trade entry and trade management. As far as trade entry is concerned, a long or short position is typically initiated based on various indicators such as moving average crossings or chart patterns such as head-and-shoulders, cup and handles and breakouts. As intimated above, the conditions surrounding trade entry are not important to our analysis. In fact, the conditions for trade entry are entirely irrelevant to our *Positive Expectation Theorem* to follow. One possible interpretation of our work is that trade management is much more important than trade entry.

Two questions which we consider in our research are as follows: Once a position has been initiated, how should one add or subtract stock over time? If the trend-following trader does not have a price model with any type of predictive power, is it possible to provide an apriori guarantee that a “favorable” outcome will result? Surprisingly, for the precise scenario which we consider in the sequel, that is, a stock price following a Geometric Brownian Motion (GBM) with a time-varying drift $\mu(t)$, it turns out that a positive expected value for the trading gain (profit) can be guaranteed. The justification of our claims involves a purely theoretical argument. To our knowledge, this paper provides the first result along these lines. A second possible interpretation of our result is that it supports the earlier papers arguing in favor of technical analysis based on statistical analysis. That is, the theory suggests that many technical analysis strategies may have an underlying rigorous justification which is yet to be discovered.

To achieve the type of results claimed above, we first describe a novel feedback-based description of trend-following strategies which we call Simultaneous Long-Short (SLS). The SLS approach is similar to other trend-following strategies in that it is direction independent and assumes no a-

³When profits are allowed to run, a loss is typically viewed differently from a drawdown. For example, if a trend-following strategy has a large profit in a current position, it may be forgiving of drawdowns from the peak of the trend. This is the “let profits run” principle in action. On the other hand, if a new position is opened and the stock initially moves against the trader at the start, losses will typically be cut short. Thus, losses in new positions are considered in a different light from drawdowns from the peak in the middle of a profitable trend.

priori model for stock price movement. We see in the sequel, that this trading strategy, motivated by feedback control considerations, has some remarkable adaptive properties which enables the trader to capture both up and down trends while cutting losses short. Perhaps the most significant aspect of this strategy is its remarkable robustness properties which are described below.

II. PRICE DYNAMICS, TRADING GAINS AND OUR CLAIMS

We consider a market with time-varying stock price $p(t)$ governed by the time-varying Geometric Brownian Motion (GBM)

$$\frac{dp}{p} = \mu(t)dt + \sigma(t)dZ_t$$

where $\mu(t)$ is the so-called *drift*, $\sigma(t)$ is the *volatility* and dZ_t is a standard Wiener process increment. In the sequel the previously stated assumptions about an idealized market are assumed throughout. Note that assumptions of this sort also pervade many of the fundamental papers on markets; e.g., see [17] and [18]. Letting $I(t)$ be the instantaneous investment, the resulting stochastic equation for the trading gain or loss $g(t)$ is given by

$$dg = \frac{dp}{p}I(t) = (\mu(t)dt + \sigma(t)dZ_t)I(t).$$

With the simple notation above and no further assumptions, the main results of this paper are easy to paraphrase: There exists a technical trend-following trading rule, namely an algorithm for varying the investment function $I(t)$, which is robust to rather arbitrary variations in $\mu(t)$ and $\sigma(t)$ and guarantees an “edge” against the market. More precisely, for continuous non-vanishing $\mu(t)$ with unknown sign and magnitude, we prove that

$$\mathbb{E}[g(t)] > 0$$

for all $t \geq 0$. That is, regardless of the parameters governing the GBM, an SLS trend follower robustly expects to make money on every trade. When we say “robustly” above, we mean that a positive expected value is invariant to a large class of admissible time variations in $\mu(t)$ and $\sigma(t)$.

Lack of Predictability: When pondering the possibility of such a result, it is important for the reader to consider the fact that the admissibility of a time-varying drift $\mu(t)$ rules out some obvious strategies which would work well in the time-invariant case. That is, in the time-invariant case, one obvious method to gain an “edge” is to estimate the sign of the drift μ over some

training period and then go long if μ is positive and short if μ is negative. Surprising as it may seem, in this paper we prove that positive expectation can be assured even when the drift μ is time-varying and can change sign rather arbitrarily. Another point to note is that the lack of differentiability of the Brownian Motion also rules out obvious strategies which amount to estimating a derivative of the price, call it $p'(t)$, and “taking a ride,” long or short, based on its sign. In the jargon of feedback theory, the SLS strategy which we describe below is said to have differentiator-free dynamics.

Crude Analogy: In view of the unknown two-signed time variations in $\mu(t)$ which makes it appear to be impossible to exploit the underlying GBM price process, as previously stated, our trading algorithm for $I(t)$ uses no “internal model” for stock prices; i.e., our strategy is model-free.⁴ Instead of using a price model, changes in $I(t)$ are performance driven. That is, we use a feedback rule which is reactive to trading gains or losses $g(t)$ rather than predictive.

To provide a crude analogy which illuminates why a reactive feedback control approach can often be powerful in a model-free context, we consider a simple scenario which is second nature to those well-versed in feedback theory: Imagine controlling the water inflow into a bathtub with the goal of attaining eighty degree temperature. In this setting, a modelling approach would be to work with thermodynamics equations of the fluid inflow to determine the appropriate setpoint for the hot and cold control levels. In contrast, a model-free feedback rule, based on sensing the water temperature T , is readily available and much simpler: If T is too low, add hot water. If T is too high, add cold water. Our view of stock trading is based on similar considerations: Thinking of the level of investment $I(t)$ as playing the role of the water inflow regulator and the gains or losses $g(t)$ as playing the role of the water temperature, our trading algorithm $I(t)$ adapts to $g(t)$ without assuming knowledge of the underlying Geometric Brownian Motion for prices.

⁴It is important not to confuse the fact that the strategy is model-free with the robustness results that show the positive expectation gain property for a class of price dynamics. The strategy is model-free in that trading decisions are not based upon any model of prices. We then provide rigorous proof that this model-free trading rule provides a robustly positive expected trading gain over the class of GBM price processes with time-varying drift and volatility.

Embellishments of Main Result: In addition to the positive expected value result described above, this paper also includes some additional embellishments on the main result. That is, we provide details about the statistics of the trading gain $g(t)$. This includes a closed-form solution for its variance and a description of its probability density function.

III. THE SIMULTANEOUS LONG-SHORT (SLS) TECHNICAL TRADING RULE

Motivated by our previous work [15] and [16] in the feedback control community involving the simpler case with time-invariant drift and volatility, we take the instantaneous investment $I(t)$ to be the sum of two positions, the long trade $I_L(t)$ and the short trade $I_S(t)$. Noting the signs $I_L(t) > 0$ and $I_S(t) < 0$, we have

$$I(t) \doteq I_L(t) + I_S(t).$$

To facilitate understanding of the trading strategy, it is convenient to temporarily view I_L and I_S as two separate trades which can be “netted out” when implemented in a brokerage account; e.g., at some time $t > 0$, if $I_L = \$7,000$ and $I_S = -2,500$, then $I(t) = \$4,500$ is the net position which is long. In summary, these two trades can be viewed as running simultaneously in parallel.

Feedback Functions: Now, to complete the description of the trading strategy, we need to specify the rules for selection of I_L and I_S . Indeed, letting $g_L(t)$ and $g_S(t)$ denote the cumulative trading gain or loss over $[0, t]$ from the long and short trades respectively, and taking $I_0 > 0$, the initial investment and $K > 0$, the so-called feedback gain, the long and short investments are given by

$$I_L(t) \doteq I_0 + Kg_L(t) \quad \text{and} \quad I_S(t) \doteq -I_0 - Kg_S(t)$$

with resulting net investment

$$\begin{aligned} I(t) &= I_L(t) + I_S(t) \\ &= K(g_L(t) - g_S(t)). \end{aligned}$$

Using the jargon of feedback control theory we describe I_L and I_S as *static linear feedbacks*. Finally, since there are no gains or losses at time zero, the condition $g_L(0) = g_S(0) = 0$ makes it apparent that where $I_0 > 0$ corresponds to the initial investment in each trade, and $K > 0$,

the feedback gain, while arbitrary in our analysis, is a measure of how “aggressively” the trade gets incremented or decremented over time. To summarize the exposition in this section, we provide the block diagram below which shows all the interconnection of feedback variables which describe our trading scheme.

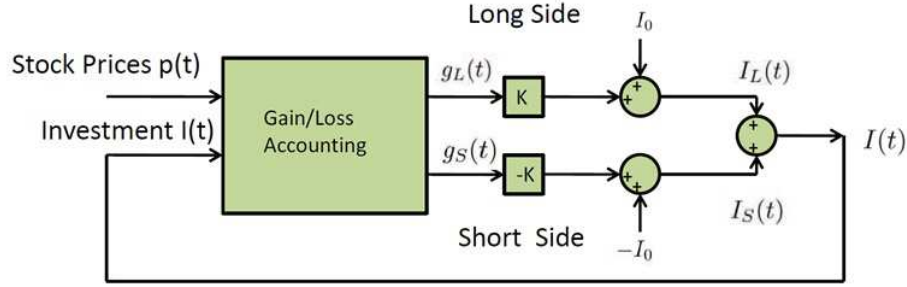


Fig. 1: Feedback Control Loop for SLS Trading

Adaptation Without a Price Model: To conclude this section, as a preamble to our positive-expectation result to follow, we make a number of simple observations which facilitate understanding the trading strategy being used. First, since the long and short trade can be viewed as running independently, the overall trading gain or loss is given by

$$g(t) = g_L(t) + g_S(t).$$

Second, at starting time $t = 0$, the initial condition $g_L(0) = g_S(0) = 0$ in combination with the formula for I_L and I_S forces $I(0) = 0$. That is, the trade is initially neutral. Finally, we note that the trading rule has “built-in” adaptive properties which are apparent from the formulae above. Namely, if one side of the trade (long or short) flourishes, the amount invested in it will increase and the amount invested on the other side will decline. That is, the overall investment will become more heavily weighted in favor of the winning side. In this sense, our strategy is a “trend follower.” To say all this more concretely, imagine a bull market. Then as the stock price goes up, our two feedback rules above guarantee that the long position I_L will be amplified and the short position I_S will be attenuated. In the next section, we prove that the net result of this adaptive process is that positive expectation of the trading gain is guaranteed.

Before going to the main result, the following point, mentioned in the introduction, is important to emphasize. The investment function $I(t)$ uses no a priori knowledge of drift $\mu(t)$ and volatility $\sigma(t)$ from the underlying Geometric Brownian Motion driving the price $p(t)$. Neither does the simple recipe for $I(t)$ include an estimator for these functions. In fact, given that $\mu(t)$ can change sign rather instantaneously, estimation of the drift is rather futile. We simply view the investment function as a “reactive” model-free controller.

IV. THE MAIN RESULT: THE POSITIVE EXPECTATION THEOREM

The following theorem, the main result of this paper, indicates that every trade made by the SLS trend-follower has a robustly positive expected gain.

Positive Expectation Theorem: *Assume the stock price $p(t)$ is governed by the Geometric Brownian Motion*

$$\frac{dp}{p} = \mu(t)dt + \sigma(t)dZ_t$$

where dZ_t is a standard Wiener process increment, and $\mu(t)$ and $\sigma(t)$ are time-varying continuous functions. Then, for $t \geq 0$, the expected gain of the SLS trading gain $g(t)$ is given by

$$\mathbb{E}[g(t)] = \frac{I_0}{K} \left[e^{K \int_0^t \mu(\tau) d\tau} + e^{-K \int_0^t \mu(\tau) d\tau} - 2 \right]$$

and satisfies $\mathbb{E}[g(t)] \geq 0$. Moreover, for all drifts satisfying

$$\int_0^t \mu(\tau) d\tau \neq 0,$$

we have

$$\mathbb{E}[g(t)] > 0.$$

Proof: We first solve for the trading gain $g_L(t)$ associated with the long investment $I_L(t)$. Under the stochastic differential equation for the price, the associated trading gain satisfies

$$dg_L = I(t) \frac{dp}{p} = (I_0 + Kg_L)(\mu(t)dt + \sigma(t)dZ)$$

whose solution is readily shown to exist and be unique.⁵ We now convert the stochastic differential equation above to a GBM via the substitution $f \doteq I_0 + Kg_L$. Indeed, this leads to the equation

$$\frac{df}{f} = K\mu(t)dt + K\sigma(t)dZ.$$

Now by Ito's Lemma, for example, see [19], when applied to $\log f$, we have

$$d(\log f) = \left[K\mu(t) - \frac{1}{2}K^2\sigma^2(t) \right] dt + K\sigma(t)dZ$$

which can be integrated and exponentiated to obtain the closed form solution

$$f(t) = e^{K \int_0^t \mu(\tau)d\tau - \frac{1}{2}K^2 \int_0^t \sigma^2(\tau)d\tau + K \int_0^t \sigma(\tau)dZ(\tau)} f(0).$$

Substituting back into $f(t) = I_0 + Kg_L(t)$ and solving for g_L leads to

$$g_L(t) = \frac{1}{K} \left(e^{K \int_0^t \mu(\tau)d\tau - \frac{1}{2}K^2 \int_0^t \sigma^2(\tau)d\tau + K \int_0^t \sigma(\tau)dZ(\tau)} - I_0 \right) [I_0 + Kg_L(0)].$$

Finally, recognizing that

$$\frac{p(t)}{p(0)} = e^{\int_0^t (\mu(\tau)d\tau - \frac{1}{2}\sigma^2(\tau))d\tau + \int_0^t \sigma(\tau)dZ(\tau)}$$

and recalling that $g_L(0) = 0$, we obtain

$$g_L(t) = \frac{I_0}{K} \left(\left(\frac{p(t)}{p(0)} \right)^K e^{\frac{1}{2}(K-K^2) \int_0^t \sigma^2(\tau)d\tau} - 1 \right).$$

The gain on the short trade is similarly obtained by replacing I_0 and K with $-I_0$ and $-K$; i.e.,

$$g_S(t) = \frac{I_0}{K} \left(\left(\frac{p(t)}{p(0)} \right)^{-K} e^{-\frac{1}{2}(K+K^2) \int_0^t \sigma^2(\tau)d\tau} - 1 \right).$$

Thus, the gain on the SLS feedback controller is given by the sum of g_L and g_S . Namely,

$$g(t) = \frac{I_0}{K} \left[\left(\frac{p(t)}{p(0)} \right)^K e^{\frac{1}{2}(K-K^2) \int_0^t \sigma^2(\tau)d\tau} + \left(\frac{p(t)}{p(0)} \right)^{-K} e^{-\frac{1}{2}(K+K^2) \int_0^t \sigma^2(\tau)d\tau} - 2 \right].$$

Before computing the expectation of this trading gain, we define the following quantities that will simplify notation

$$a \doteq e^{\frac{1}{2}(K-K^2) \int_0^t \sigma^2(\tau)d\tau}; \quad c \doteq e^{-\frac{1}{2}(K+K^2) \int_0^t \sigma^2(\tau)d\tau}.$$

⁵Note that the existence and uniqueness of the solution to the stochastic differential equation for $g_L(t)$ on any finite time interval $t \in [0, T]$ follows from the continuity of $\mu(t)$ and $\sigma(t)$. More specifically, since the continuous functions $\mu(t)$ and $\sigma(t)$ are bounded on any closed interval $[0, T]$, one may easily verify that the drift term $(I_0 + Kg_L)\mu(t)$ and diffusion term $(I_0 + Kg_L)\sigma(t)$ satisfy the requisite Lipshitz conditions; see [19], Theorem 5.2.1.

Now, recalling the fact that the K -th moment of a log normal random variable X denoted by $\log X \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ is

$$\mathbb{E}[X^K] = e^{K\mu_Y + \frac{1}{2}K^2\sigma_Y^2},$$

by taking $X \doteq \frac{p(t)}{p(0)}$; $\mu_Y \doteq \int_0^t (\mu(\tau) - \frac{1}{2}\sigma^2(\tau))d\tau$; $\sigma_Y \doteq \int_0^t \sigma^2(\tau)d\tau$, the expected value of the trading gain is given by

$$\begin{aligned} \mathbb{E}[g(t)] &= \frac{I_0}{K} (a\mathbb{E}(X^K) + c\mathbb{E}(X^{-K}) - 2) \\ &= \frac{I_0}{K} \left(ae^{K\mu_Y + \frac{1}{2}K^2\sigma_Y^2} + ce^{-K\mu_Y + \frac{1}{2}K^2\sigma_Y^2} - 2 \right). \end{aligned}$$

Then, upon substitution for μ_Y, σ_Y, a and c , we obtain the formula for $\mathbb{E}[g(t)]$. Now, to complete the proof, we simply note that $\mathbb{E}[g(t)] > 0$ follows immediately from the fact that the function $e^x + e^{-x} - 2$ everywhere non-negative and strictly positive for $x \neq 0$. $\square$

Remarks: Note the astonishing robustness of the SLS trading rule. Except for the “singular” break-even case obtained when

$$\int_0^t \mu(\tau)d\tau = 0,$$

the expected value of the resulting trading gain is positive. That is, one expects a positive trading gain by following the SLS trading rule, regardless of the actual drift and volatility functions, $\mu(t)$ and $\sigma(t)$. Said another way, as long as $\mu(t)$ results in a “trend” by achieving non-zero $\int_0^t \mu(\tau)d\tau$, the SLS trader expects to make a profit. To the best of our knowledge, this result is the first theoretically-based justification of why technical trend-following trade-management rules may be expected to work.

V. EMBELLISHMENTS ON SLS TRADING STRATEGY

In this section, we provide a number of additional results which are straightforward to obtain via calculations which begin with the sample path solution

$$g(t) = \frac{I_0}{K} \left[\left(\frac{p(t)}{p(0)} \right)^K e^{\frac{1}{2}(K-K^2) \int_0^t \sigma^2(\tau)d\tau} + \left(\frac{p(t)}{p(0)} \right)^{-K} e^{-\frac{1}{2}(K+K^2) \int_0^t \sigma^2(\tau)d\tau} - 2 \right]$$

given in the proof of the theorem above.

Variance of Trading Gain: Combining the formula above with the fact that $p(t)$ is lognormally distributed, a lengthy but straightforward calculation leads to

$$Var[g(t)] = \frac{I_0^2}{K^2} \left(e^{K^2 \int_0^t \sigma^2(\tau) d\tau} - 1 \right) \left(e^{2K \int_0^t \mu(\tau) d\tau} + e^{-2K \int_0^t \mu(\tau) d\tau} + e^{-K^2 \int_0^t \sigma^2(\tau) d\tau} \right).$$

Worst-Case Performance: It is easy to show that the support for the probability density function for $g(t)$ is bounded on the left. Via a lengthy calculation, it can be shown that the worst-case trading loss is given by

$$g_W(t) = \frac{2I_0}{K} \left[e^{-\frac{1}{2}K^2 \int_0^t \sigma^2(\tau) d\tau} - 1 \right].$$

That is, it is guaranteed that $g(t) \geq g_W(t)$ with probability one. Consistent with the maxims of trend trading, this represents a rigorous argument that losses can be cut in a provable way under the SLS trading rule.

The Probability Density Function: Via lengthy calculations which take off from the SLS sample path solution for $g(t)$, it is actually possible to obtain the entire probability density function for the gains and losses $g(t)$ as a closed form expression. Putting aside the trivial degenerate case where $\sigma(t) \equiv 0$, the calculations are summarized as follows: Let

$$\begin{aligned} X_{\pm}(x, t) &\doteq \frac{1}{2} \left[\left(\frac{K}{I_0}x + 2 \right) \pm \sqrt{\left(\frac{K}{I_0}x + 2 \right)^2 - 4e^{-K^2 \int_0^t \sigma^2(\tau) d\tau}} \right] e^{\frac{1}{2}(K-K^2) \int_0^t \sigma^2(\tau) d\tau}; \\ Z_{\pm}(x, t) &\doteq \frac{\left[\log X_{\pm}^{\frac{1}{K}}(x, t) - \int_0^t (\mu(\tau) - \frac{1}{2}\sigma^2(\tau)) d\tau \right]^2}{\int_0^t \sigma^2(\tau) d\tau}; \\ A(x, t) &\doteq \frac{1}{I_0 \sqrt{2\pi \int_0^t \sigma^2(\tau) d\tau} \left[\left(\frac{K}{I_0}x + 2 \right)^2 - 4e^{-K^2 \int_0^t \sigma^2(\tau) d\tau} \right]}. \end{aligned}$$

Then, for $t > 0$, the probability density function $f(x, t)$ for the trading gain $g(t)$ is as follows: For $x \leq g_W(t)$, consistent with the worst-case analysis, $f(x, t) \equiv 0$. Moreover, for $x > g_W(t)$,

$$f(x, t) = A(x, t) \left(e^{-\frac{1}{2}Z_+^2(x, t)} + e^{-\frac{1}{2}Z_-^2(x, t)} \right).$$

Numerical Example: The probability density $f(x, t)$ above, can readily be plotted. For example, in Figure 2, we provide illustrative plots with $I_0 = 1$, $K = 4$, $t = 0.5$, $\sigma = 0.2$, and constant μ .

One salient feature of these plots, consistent with common sense, is that the larger the realized value of the ratio $\gamma \doteq \frac{|\mu|}{\sigma}$, the more attractive the trade becomes. For example, in the figure, for those plots when $\gamma > 2$, it is obvious by inspection that the probability of a significant rate of return is quite high; e.g., when $|\mu| = 1$, by performing a straightforward integration, we conclude that the probability that the raw rate of return is 100% is about 0.55. For this same scenario, the probability of a 25% return is about 0.94.

On the other hand, when the realized value of the γ ratio is small, the likelihood of a loss can be quite high. For example, for the “driftless” case corresponding to $\gamma = 0$, a straightforward integration of the density function results in a probability $p \approx 0.62$ that $g(0.5) < 0$ and a probability $p \approx 0.2$ that a raw return of 10% or more will result. Finally, to provide a balanced picture of this low γ trade, it is important to recall one important aspect of the theorem: That is, for the SLS trade, the loss is no more than $g_W(t)$ with probability one. For this scenario under consideration, we obtain $g_W(0.5) \approx -0.07$. That is, a loss of about 7% is the maximum that can occur. Many other possibilities exist to explore the properties of the SLS trade via the

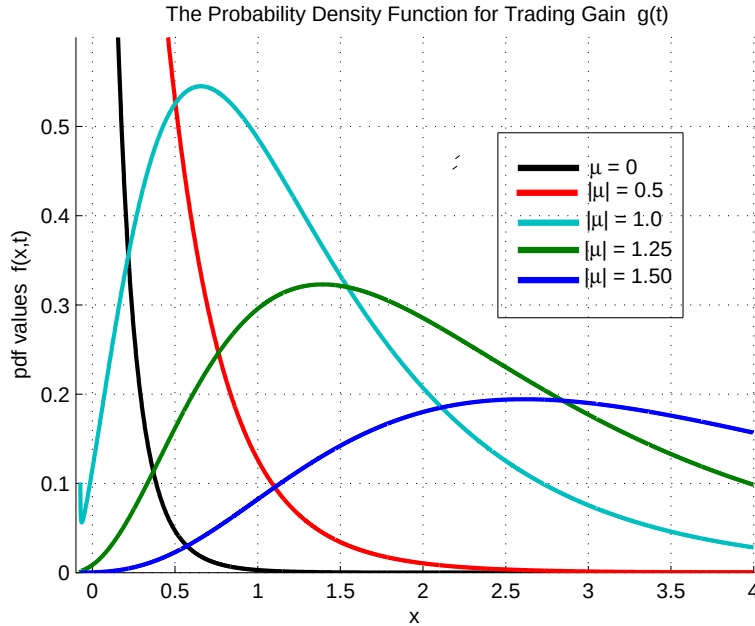


Fig. 2: Probability Density Function for $g(t)$ with $\sigma = 0.2$, $K = 4$, $t = 0.5$ and $I_0 = 1$.

development of formulae derived from the solution for $g(t)$.

VI. CONCLUSION

This paper provide a new theoretically-based argument as to the validity of model-free trend-following strategies. We argue that the “secret sauce” of trend-following rules comes from the management of the trade as manifested via the reactive nature of feedback. We define the feedback-based Simultaneous-Long-Short (SLS) trend-following rule, and rigorously prove that it achieves remarkable levels of robustness without the use of any predictive model for price dynamics. In fact, our results suggest that rather than “betting the farm” on the predictive power of technical signals, the trend follower does just the opposite and renders the trade as robust as possible to incorrect predictions, so much so that one still expects to profit (although not necessarily to the same degree) even if the underlying price process is vastly different from what may have been predicted. Thus, what academia has long dismissed as “voodoo” may in fact represent a different paradigm for trading that emphasizes feedback-induced robustness over predictability of price models.

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